ITA06020 MACHINE LEARNING

Experiment 14 – House Price Prediction Using Machine Learning

Aim:

To implement a House Price Prediction model using a suitable Machine Learning algorithm in Python to predict house prices based on housing features.

Dataset: California Housing Dataset (real-world dataset available in Scikit-learn).

Algorithm: Linear Regression.

Procedure

1. Import the required Python libraries.
2. Load the California Housing dataset.
3. Separate input features and target values.
4. Split the dataset into training and testing data.
5. Create a Linear Regression model.
6. Train the model using training data.
7. Predict house prices using testing data.
8. Evaluate the model using Mean Squared Error and R^2 score.

Python Program

```
from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

data = fetch_california_housing()

X = data.data

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42

)

model = LinearRegression()

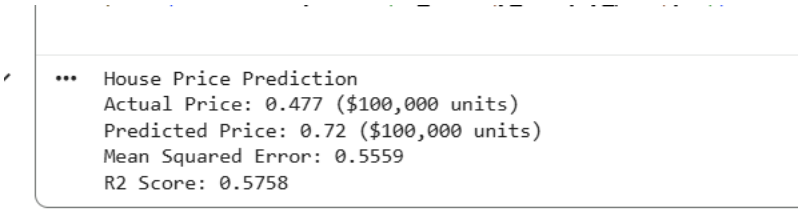
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("House Price Prediction")
```

```
print("Actual Price:", y_test[0], "($100,000 units)")
print("Predicted Price:", round(y_pred[0], 2), "($100,000 units)")
print("Mean Squared Error:",
    round(mean_squared_error(y_test, y_pred), 4))
print("R2 Score:", round(r2_score(y_test, y_pred), 4))
```

Output:



Result

Thus, the House Price Prediction model was successfully implemented using Linear Regression and the California Housing dataset. The model performance was evaluated using Mean Squared Error and R^2 score.